

Temperature Linear Regression Model Report

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Dataset Summary

Measure	Value
Rows	3,655
Columns	6
Cities	5
Date range	2023-01-01T00:00:00 to 2024-12-31T00:00:00
Split	2920 train / 735 test

Train Metrics

Metric	Value	Confidence interval
MAE	2.6977	2.5124 to 2.8671
RMSE	3.4626	3.1854 to 3.6952
MAPE	50.9675	29.5716 to 72.2106
R2	0.8744	0.8514 to 0.8893

Test Metrics

Metric	Value	Confidence interval
MAE	3.0958	2.6884 to 3.5974
RMSE	4.0643	3.4971 to 4.6973
MAPE	84.7329	29.2945 to 163.5565
R2	0.8167	0.7753 to 0.8416

Model Configuration

Setting	Value
random_seed	42
test_size	0.2
shuffle	False
normalize_features	True
feature_selection_mode	all
selected_cities	None
target_column	Temperature_C
date_column	Date_Time
city_column	City
temporal_features	['day_of_year', 'month', 'year', 'day_of_week']

Top Feature Coefficients

Feature	Coefficient	Absolute
city__City_Bandar_Abbas	7.8016	7.8016
temporal__day_of_year_cos	-7.7209	7.7209
city__City_Mashhad	-3.4296	3.4296
city__City_Sanandaj	-3.1569	3.1569
temporal__day_of_year_sin	-2.8260	2.8260
city__City_Rasht	-2.7961	2.7961
city__City_Yazd	1.5810	1.5810
temporal__month	0.7951	0.7951
temporal__day_of_year	-0.6826	0.6826
temporal__month_sin	0.2867	0.2867
temporal__year	0.2426	0.2426
temporal__month_cos	0.1600	0.1600
temporal__day_of_week_cos	0.0505	0.0505
temporal__day_of_week_sin	0.0297	0.0297
temporal__day_of_week	-0.0163	0.0163

Magnitude is affected by preprocessing. Numeric temporal features are standardized when `normalize_features` is enabled.